CONSTRUCTION OF THE CANONICAL FUNCTORIAL MAP FROM THE ALTERNATING CECH COMPLEX OF A FINITE COVERING TO DERIVED GLOBAL SECTIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites six cross-chapter labels (including a theorem), each recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The author's closing sentence omits only the verification of functoriality and of the compatibilities of the constructed map; that sentence is kept verbatim in the proof and the rest of the argument is complete.

Allowed standard prerequisites (editorial).

- Existence of a K-injective embedding. Stacks Project, Injectives, Theorem injectives-theorem-K-injective-embedding-grothendieck (tag 079P). Supplies the K-injective representative of F^bullet used to define the map.
- Comparison of the alternating and usual Cech complexes and injective triviality. Stacks Project, Cohomology of Sheaves, Lemmas lemma-injective-trivial-cech (tag 08C2) and lemma-alternating-usual (tag 01FM). Provide the comparison that turns the levelwise map into the required map into derived global sections.

Source reference. Original equation 08C0: sources/stacks-chow/tags.txt, source lines 10585–10585. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0133: sources/stacks-chow/tags.txt, source lines 1421–1421. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01EP: sources/stacks-chow/tags.txt, source lines 1842–1842. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01FM: sources/stacks-chow/tags.txt, source lines 1875–1875. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 079P: sources/stacks-chow/tags.txt, source lines 9279–9279. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{U} : X = \bigcup_{i \in I} U_i$ be a finite open covering. For a complex $\mathcal{F}^\bullet$ of $\mathcal{O}_X$-modules there is a canonical map*

$$Tot(\check{\mathcal{C}}^\bullet_{alt}(\mathcal{U}, \mathcal{F}^\bullet)) \longrightarrow R\Gamma(X, \mathcal{F}^\bullet)$$

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functorial in $\mathcal{F}^\bullet$ and compatible with (08C0).

Proof. Let $\mathcal{I}^\bullet$ be a K-injective complex whose terms are injective $\mathcal{O}_X$-modules. The map (08C0) for $\mathcal{I}^\bullet$ is a map $\Gamma(X, \mathcal{I}^\bullet) \to \mathrm{Tot}(\check{\mathcal{C}}^\bullet_{alt}(\mathcal{U}, \mathcal{I}^\bullet))$. This is a quasi-isomorphism of complexes of abelian groups as follows from Homology, Lemma 0133 applied to the double complex $\check{\mathcal{C}}^\bullet_{alt}(\mathcal{U}, \mathcal{I}^\bullet)$ using Lemmas 01EP and 01FM. Suppose $\mathcal{F}^\bullet \to \mathcal{I}^\bullet$ is a quasi-isomorphism of $\mathcal{F}^\bullet$ into a K-injective complex whose terms are injectives (Injectives, Theorem 079P). Since $R\Gamma(X, \mathcal{F}^\bullet)$ is represented by the complex $\Gamma(X, \mathcal{I}^\bullet)$ we obtain the map of the lemma using

$$\mathrm{Tot}(\check{\mathcal{C}}^\bullet_{alt}(\mathcal{U}, \mathcal{F}^\bullet)) \longrightarrow \mathrm{Tot}(\check{\mathcal{C}}^\bullet_{alt}(\mathcal{U}, \mathcal{I}^\bullet)).$$

We omit the verification of functoriality and compatibilities. □